

TASHKENT INTERNATIONAL UNIVERSITY · FACULTY OF ECONOMICS Joint distributions, covariance and correlation

Week 6 · Lectures 1 and 2

Tashkent International University · Faculty of Economics

2025/2026 academic
year



From one variable to two

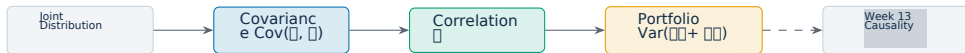
Weeks 4-5. One random variable: $E[\cdot]$, $\text{Var}(\cdot)$, distribution, model selection.

A common mistake But economic questions are almost always about two variables.

Budget and income. Inflation and unemployment. Two assets together.

This week's question How do two random variables behave together — and what decisions does that require?

The answer boils down to one number: correlation. But that number hides a lot — as we'll see.



Learning Objectives for Week 6

By the end of this week, you will:

1. Construct joint and marginal distributions.
2. Verify the independence of random variables.
3. Derive and calculate covariance from the definition.
4. Interpret correlation and show its limitations.
5. Apply $\text{Var}(X + Y)$ and quantify diversification.
6. Show from data that correlation does not imply causation.

Learning Outcome CLO4. Analyze joint distributions, covariance, and correlation (Bloom: Analysis).

Common Error This week Control Work 2. Material from weeks 4–6: random variables, discrete and continuous distributions, covariance and correlation.

Joint distribution

Definition Joint Distribution

$$P(\Omega, \Omega) = P(\Omega = \Omega, \Omega = \Omega)$$

Marginal Distribution

$$p_X(x) = \sum_y p(x, y)$$

Marginal distributions are sums written on the margins of a table. Hence the name.

Uzbekistan is above the median; Kazakhstan is above the median ($\Omega = 27$)

	$y = 0$	$y = 1$	$p_X(x)$
$x = 0$	0,3333	0,1852	0,5185
$x = 1$	0,1852	0,2963	0,4815
$p_Y(y)$	0,5185	0,4815	1,0000

A common mistake Marginal distributions do not restore jointness. There are joint tables with the same margins, but completely different ones.

Independence of random variables

Definition X and Y are called independent if for all x, y

$$p(x, y) = p_X(x) p_Y(y)$$

Equivalent: $P(Y=y | X=x) = P(Y=y)$.

A common mistake is “For all x, y ” — important. The fact that equality holds in one cell does not prove independence; its violation in one cell refutes it.

We check ($X=1, Y=1$):

$$p(1, 1) = 0.2963$$

$$P(X=1) P(Y=1) = 0.4815 \times 0.4815 = 0.2318$$

$$0.2963 \neq 0.2318 \Rightarrow \text{related}$$

In conditional form:

$$P(X=1 | Y=1) = \frac{0.2963}{0.4815} = 0.6154$$

$$\text{vs } P(X=1) = 0.4815$$

In years when the harvest in Uzbekistan is good, the probability of a good harvest in Kazakhstan also increases — general weather conditions.

Covariance

Covariance: a measure of covariance

Definition Covariance

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

Calculation formula:

$$\begin{aligned}\text{Cov}(X, Y) &= E[XY - \mu_X Y - \mu_Y X + \mu_X \mu_Y] \\ &= E[XY] - \mu_X \mu_Y - \mu_Y \mu_X + \mu_X \mu_Y \\ &= E[XY] - E[X] E[Y]\end{aligned}$$

For the sample: $\hat{\text{Cov}} = \frac{1}{n-1} \sum (x_i - \bar{x})(y_i - \bar{y})$

Meaning of the notation. If both are above the mean (or both are below), the product is positive.

If one is above and the other is below — negative.

Theorem X and Y are independent $\Rightarrow \text{Cov}(X, Y) = 0$.

The converse is false: even if $\text{Cov} = 0$, there can be a relationship — just not linear.

Let's calculate by hand Uzbekistan
2010-2018: $\bar{x}$ = wheat, $\bar{y}$ = barley (t/ha)

Year	$\bar{x}$	$\bar{y}$	$\bar{x} - \bar{x}$	$\bar{y} - \bar{y}$	product
2010	4,600	1,914	-0,005	+0,391	-0,0021
2011	4,556	1,282	-0,049	-0,241	+0,0119
2012	4,710	1,658	+0,105	+0,135	+0,0141
2013	4,739	1,496	+0,134	-0,027	-0,0036
2014	4,782	1,514	+0,177	-0,009	-0,0016
2015	4,818	1,631	+0,213	+0,108	+0,0229
2016	4,802	1,625	+0,197	+0,102	+0,0200
2017	4,316	1,349	-0,289	-0,174	+0,0504
2018	4,126	1,239	-0,479	-0,284	+0,1362
$\bar{x}=4,6054$		$\bar{y}=1,5231$		0,2482	

$$s_{xy} = \frac{0,2482}{9 - 1} = \mathbf{0,03103}$$

$$s_x = 0,23976, \quad s_y = 0,21301$$

$$r = \frac{0,03103}{0,23976 \times 0,21301} = \mathbf{0,6076}$$

$$r^2 = \mathbf{0,3691}$$

2017 and 2018 — drought years — made the largest contribution (0.0504 and 0.1362). Both crops declined simultaneously.

Correlation

A common mistake The problem with covariance: unity. In the above example, the unit of $\text{Cov}(Y, X)$ is $(\text{t/ha})^2$. For budget and income, it is $(\text{million \$})^2$.

The magnitude cannot be compared.

Definition Correlation

$$\rho = \frac{\text{Cov}(X, Y)}{\text{SD}(X) \text{SD}(Y)}$$

This is the standardized covariance.

Properties:

- ▶ $-1 \leq \rho \leq 1$ is always
- ▶ unitless
- ▶ insensitive to scale
- ▶ $|\rho| = 1 \Leftrightarrow$ exactly linear relationship
- ▶ symmetric: $\rho_{XY} = \rho_{YX}$

A common error is that ρ measures only linear relationships. In strong but skewed relationships, ρ can be close to zero.

Correlation matrix: five-country wheat yield, 1992-2018

	UZB	KYRG	KAZ	TAJ	TURK
Uzbekistan	1.000	-0.096	0.407	0.917	-0.213
Kyrgyzstan	1.000	0.343	0.012	0.156	Kazakhstan
1.000	0.399	-0.133	Tajikistan	1.000	-0.392
Turkmenistan	1.000				

Strongest: Uzbekistan-Tajikistan $\rho = 0.917$. Adjacent, similar irrigation system.

Common error Most negative: Tajikistan-Turkmenistan $\rho = -0.392$. This is the most valuable pair for diversification — we'll see later.

Audience Question $\rho = 0.917$ doesn't mean Uzbekistan affects Tajikistan. What does it mean? We'll see the answer in the last section.

"Correlation" is one word, three different numbers

Movie budget and worldwide revenue, $n = 2,467$ (real data)

Measure	Value	What it measures
Pearson	0.6995	linear correlation strength
Pearson	0.5961	(on log scale) linearity on log scale
Spearman	0.7340	monotonic correlation (by color)
Kendall	0.5478	proportion of matched pairs

Spearman is higher than Pearson ($0.734 > 0.700$): the correlation is monotonic, but not perfectly linear.

A common mistake is that the sentence "Correlation 0.70" is incomplete. Which measure? On which scale? Write this in your report.

Variance of the population

Theorem

$$\text{Var}(\square\square + \square\square) = \square^2\text{Var}(\square) + \square^2\text{Var}(\square) + 2\square\square\text{Cov}(\square, \square)$$

If independent, the last term disappears:

$$\text{Var}(\square + \square) = \text{Var}(\square) + \text{Var}(\square)$$

Common mistake Attention: $\text{Var}(\square - \square) = \text{Var}(\square) + \text{Var}(\square) - 2\text{Cov}(\square, \square)$. The variances are also added in the difference.

Numerical check ($\square = \text{Uzbekistan}$, $\square = \text{Kazakhstan}$, $\square = \square = 0.5$):

$$\text{Var}(\square) = 1.3912 \quad \text{Var}(\square) = 0.0676$$

$$\text{Cov}(\square, \square) = 0.1248$$

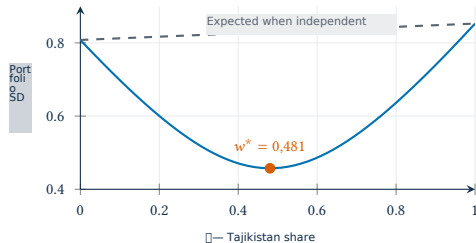
$$\text{Var}(0.5\square + 0.5\square) = 0.4271 \text{ (direct)}$$

$$0,3478 + 0,0169 + 0,0624 = 0,4271 \checkmark$$

Common error Assuming independence: $0.3647 - 14.6\%$ underestimate.

Portfolio and diversification

Diversification in numbers Tajikistan and Turkmenistan: $\rho = -0.392$



SD (Tajikistan only) 0.8528 SD
(Turkmenistan only) 0.8080 SD
(minimum portfolio) 0.4574

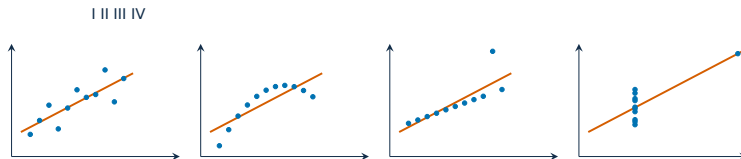
$w^* = 0.4806$ Profit 43.4%

$w^* = \frac{\text{Var}(\rho) - \text{Cov} \text{Var}(\rho) + \text{Var}(\rho) - 2 \text{Cov}}{\text{Cov} = 0.481}$ —
Almost equal share.

Assuming the common error $\rho = 0$, the portfolio SD is 0.5874 — instead of the true 0.4582 (28.2% overestimation). But note: this is in ρ levels, and there is a trend in both lines — that is what the case is testing.

What does correlation hide?

Anscombe Quartet



The set

I	9,000	3,317	7,501	2,032	0,8164	II	9,000
	3,317	7,501	2,032	0,8162	III	9,000	3,317
	7,500	2,030	0,8163	IV	9,000	3,317	7,501
	2,031	0,8165					

What Anscombe teaches

Their graphs are completely different.

I True linear relationship II A definite skewed relationship — $\bar{r}$ doesn't see it III One outlier has bent the line IV One point has created the entire relationship

Rule. Draw a scatterplot before calculating $\bar{r}$. Always.

Anscombe (1973) created these sets for exactly this purpose: to show that numerical measures are no substitute for graphs.

Audience question In set IV, 10 of the 11 points have the same $\bar{r}$. What does $\bar{r}$ measure in this kind of data?

False correlation: in real data

Uzbekistan and Tajikistan wheat yields:

$$r = 0.9167$$

Very strong. But one does not affect the other.

Common cause — time:

$$\rho(\text{year}, \text{Uzbekistan}) = 0.9076$$

$$\rho(\text{year}, \text{Tajikistan}) = 0.9624$$

Both grew in the same direction after independence.

We remove the trend. We subtract the linear trend from each row and calculate the correlation of the residuals: $\rho_{\text{residual}} = 0.3786$.

The standard error was 0.917 $\rightarrow$ 0.379. Most of the correlation was from the general trend.

The remaining 0.379 is probably a true common factor (weather, regional conditions).

Bridge to Week 13 Two time series that are growing in the same direction almost always show a strong correlation — the most common form of confounder.

**Summar
y**

Common student mistakes and misconceptions

Misconception Why it is wrong

“Cov = 0 means there is no correlation.” Only the opposite is true. Cov = 0 means there is no linear relationship; there may be a skewed relationship (Anscombe II).

“The covariance is large — the relationship is strong.” Covariance has unity. 3,816 (\$ million)² seems large, but $\rho = 0.70$. Only ρ can be compared.

“ $\rho = 0.92$ — so one affects the other.” After removing the trend, it dropped to 0.38. The overall time trend raised both lines.

“ $\text{Var}(\bar{x} - \bar{y}) = \text{Var}(\bar{x}) - \text{Var}(\bar{y})$.” $\text{Var}(\bar{x} - \bar{y}) = \text{Var}(\bar{x}) + \text{Var}(\bar{y}) - 2 \text{Cov}$. Variances are additive, not subtractive.

“It is safe to assume that the portfolio is independent.” In our example, this overestimated the SD by 28.2%; in the other direction, it may underestimate risk.

“ ρ is sufficient to describe the data.” In Anscombe’s four sets, ρ is the same, but the graphs are completely different. Draw first, then calculate.

“The marginal distributions are joint.” It is not. There are tables with the same marginals but different correlations.

Key conclusions

1. The joint distribution does not recover from the edges — the connection is additional information.

2. $\text{Cov} = E[\bar{X}\bar{Y}] - E[\bar{X}]E[\bar{Y}]$; the sign gives the direction, and the magnitude gives the dependence on unity.

3. ρ — standardized covariance; measures only linear association.

4. Don't forget the $2\rho\sigma_X\sigma_Y$ term in $\text{Var}(\bar{X} + \bar{Y})$ — a 14.6% difference in our example.

5. Negative correlation is the strongest source of diversification (43.4% gain).

6. Draw first, then calculate. Anscombe and spurious correlation prove this.

Next week is Week 7 — Sampling Distributions and the Central Limit Theorem. Finally, the $\sqrt{n}$ -law is fully explained and we move on to statistical inference.

This week's assignments

Assignment Content Due Date

CONTROL WORK 2 Graded. Weeks 4-6, 45 minutes this week Required reading Newbold 4.4-4.7 and 5.7; McClave Chapter 6; Heumann Chapter 9 Before Lecture 1 Lab W06-lab — covariance, correlation matrix, portfolio, Anscombe In the Lab

Homework W06-homework — 6 problems, 40 points Week 7, Monday Problem Set 4 MT4 Due this week Project Phase 6 Descriptive Analysis: Center, Dispersion, Shape Week 7, Friday SI Assignment Find the correlation errors of SI Week 7, Friday

Common Error Preparation for Control Work 2. You are asked to check the conditions, not memorize formulas: which distribution, why, and what happens if the condition is violated.

1. Newbold, P., Carlson, W. & Thorne, B. Statistics for Business and Economics. Pearson. 4.4–4.7 (joint distributions, covariance, portfolio) and 5.7.
2. McClave, J. T. & Sincich, T. (2018). Statistics, 13th ed. Pearson. Chapter 6.
3. Heumann, C., Schomaker, M. & Shalabh (2016). Introduction to Statistics and Data Analysis. Springer. Chapter 9.
4. Anscombe, F. J. (1973). Graphs in Statistical Analysis. The American Statistician, 27(1), 17–21. — the original source of the quartet, only 5 pages.
5. Markowitz, H. (1952). Portfolio Selection. The Journal of Finance, 7(1), 77–91. — the basis of diversification theory.
6. Matejka, J. & Fitzmaurice, G. (2017). Same Stats, Different Graphs. CHI '17 — A modern follow-up to Anscombe (“Datasaurus”).
7. Interactive: Guess the Correlation game; Seeing Theory, “Regression Analysis” section.
8. Video: StatQuest, “Covariance and Correlation, Clearly Explained”.